

Exploratory Data Analysis Report: Automobile Dataset

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Course / Track: Data Science & Software Engineering

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1. Executive Summary

This report performs an Exploratory Data Analysis (EDA) on the Automobile Dataset to evaluate structural factors influencing vehicle valuation, engine displacement, and fuel efficiency performance. The primary goal is to uncover key correlation drivers—such as horsepower, body style, and engine size—that determine price distribution across automobile manufacturers. Key findings indicate that vehicle weight and engine displacement negatively correlate with fuel efficiency, while luxury and performance categories drive high price variance.

2. Introduction & Problem Statement

Understanding vehicle pricing mechanics and fuel economy trade-offs is essential for automotive market analysis. This dataset encompasses technical attributes, structural specifications, dimensions, fuel economy ratings, and pricing across several car manufacturers.

Objectives

- Perform rigorous data cleaning and handling of missing values.
- Identify core features that correlate with vehicle price.
- Analyze manufacturer trends regarding fuel economy and power metrics.

3. Data Cleaning & Preprocessing Methodology

To prepare the dataset for statistical evaluation, several data sanitization steps were performed:

1. **Feature Reduction:** Redundant columns normalized-losses and symboling were dropped to streamline analysis.
2. **Deduplication:** Duplicate entries were identified and removed.
3. **Missing Value Processing:** Placeholder strings (?) were converted to np.nan. Rows containing missing values were removed using `.dropna()`, leaving **159 complete records** for analysis.
4. **Type Casting:**
 - Discrete numerical features (horsepower, peak-rpm, price, engine-size, city-mpg, highway-mpg, curb-weight) were converted to integer data types (int64).
 - Dimensional continuous attributes (wheel-base, length, width, height, bore, stroke, compression-ratio) were converted to floating-point numbers (float64).

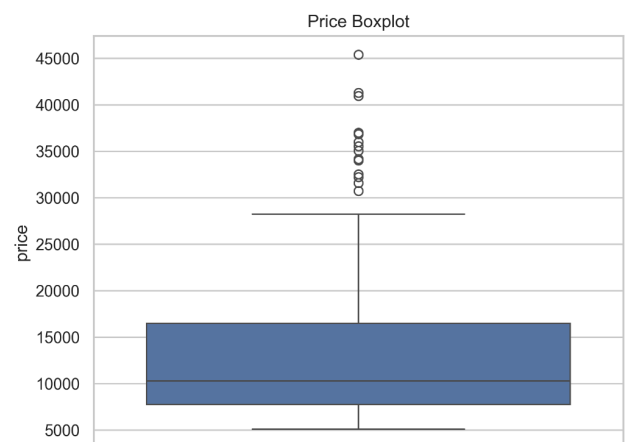
Cleaning Pipeline (Pseudo-Code)

```
# Import required libraries
# import pandas as pd
# import numpy as np
#
# Load dataset and drop redundant features
# df = pd.read_csv('automobile.csv')
# df.drop(columns=['normalized-losses', 'symboling'], inplace=True)
#
# Remove duplicates and normalize missing value markers
# df.drop_duplicates(inplace=True)
# df.replace('?', np.nan, inplace=True)
# df.dropna(inplace=True)
#
# Cast numeric types
# int_cols = ['horsepower', 'peak-rpm', 'price', 'engine-size',
# 'city-mpg', 'highway-mpg', 'curb-weight']
# df[int_cols] = df[int_cols].astype(int)
```

4. Exploratory Data Analysis & Visualizations

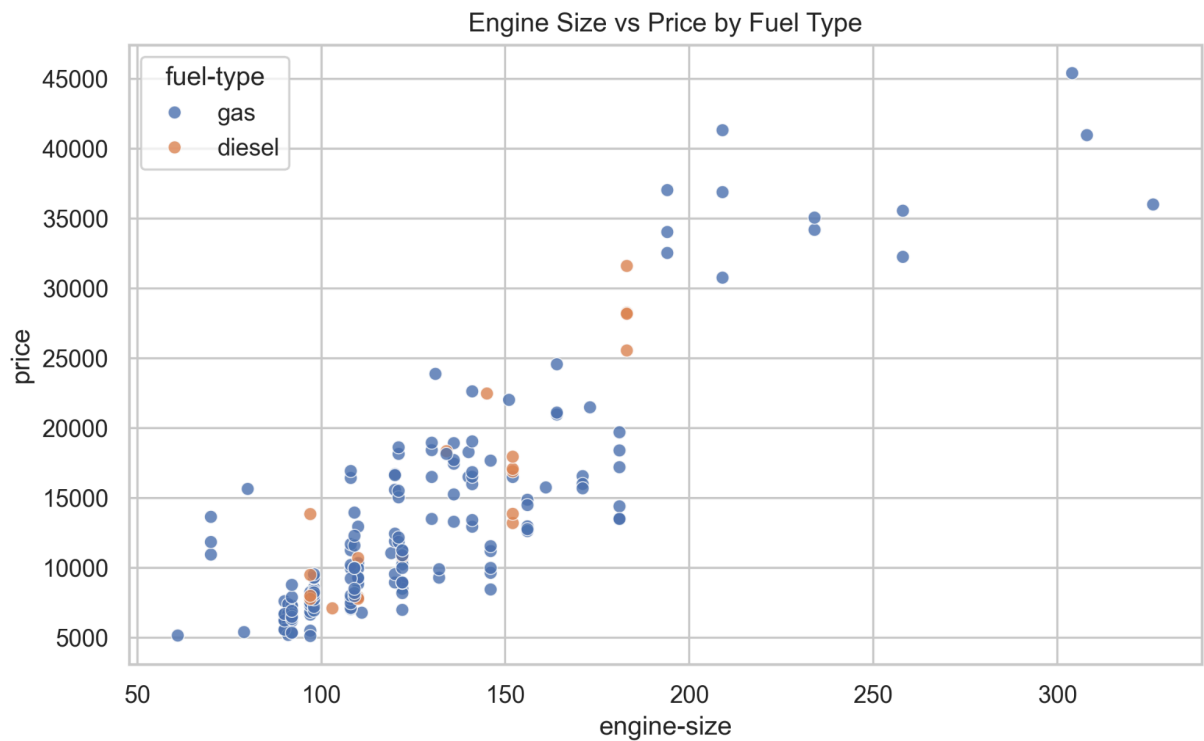
4.1 Price Distribution Analysis

- **Method:** Generated combined histogram and boxplot for the price variable.
- **Findings:** Vehicle prices are strongly right-skewed. The bulk of vehicles reside within the entry-to-mid range (\$6,000 to \$15,000), whereas luxury and specialty performance models form a long right tail stretching past \$40,000.



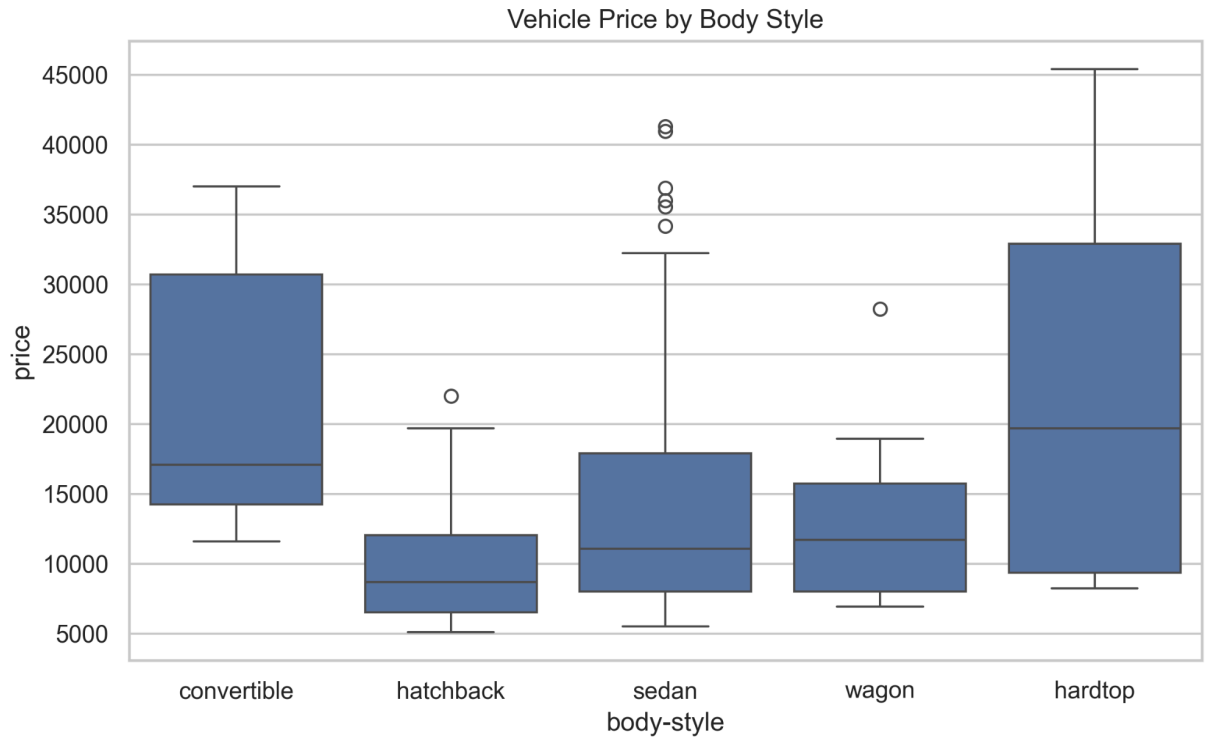
4.2 Key Price Drivers: Engine Size & Horsepower

- **Method:** Scatter plot matrix of engine-size vs. price and horsepower vs. price, categorized by fuel-type.
- **Findings:** Engine displacement and horsepower show a strong positive linear relationship with vehicle price. Gas vehicles span the full price spectrum, whereas diesel options are concentrated in the mid-to-high pricing tiers.



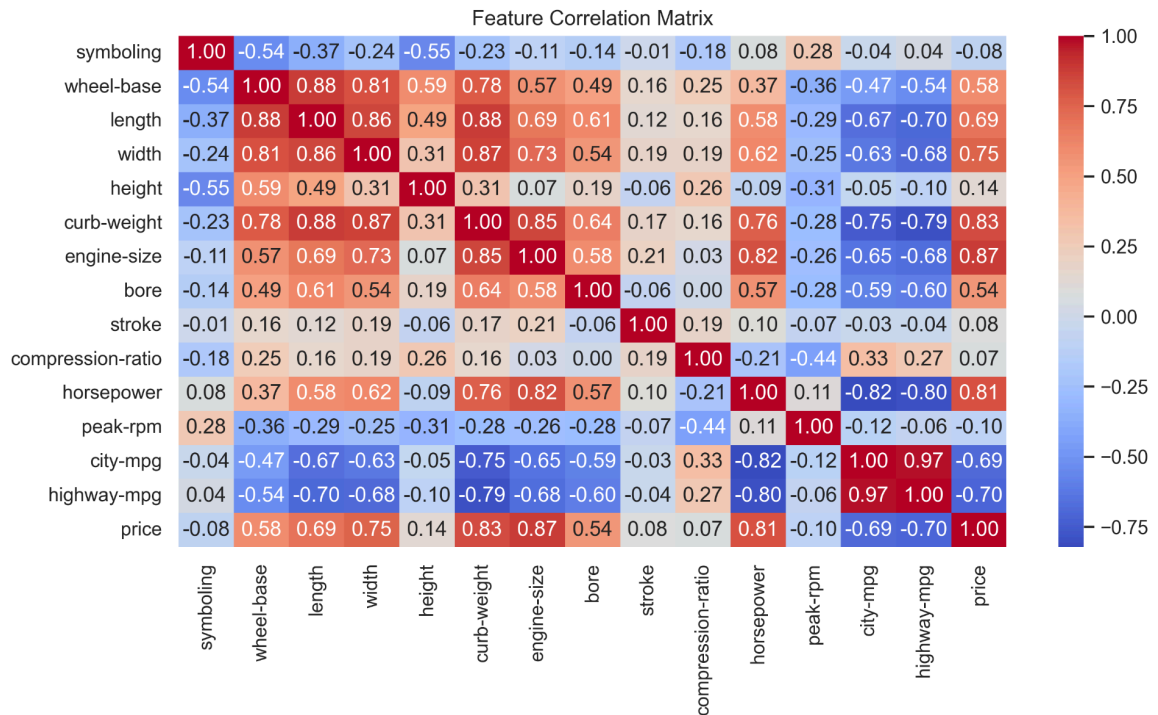
4.3 Vehicle Price by Body Style

- **Method:** Category boxplots comparing price across body-style categories.
- **Findings:** Convertible and hardtop styles command the highest median market prices, while hatchbacks represent the most budget-conscious category.



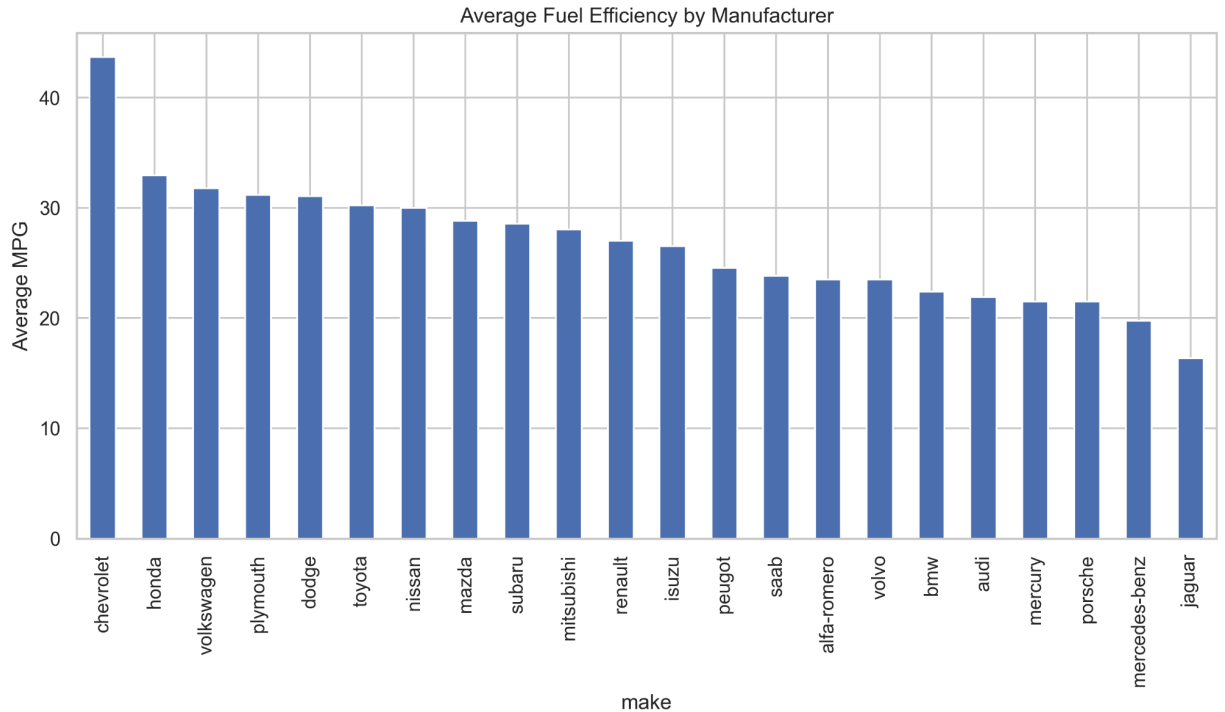
4.4 Feature Correlation Matrix

- **Method:** Seaborn heatmap computed using Pearson correlation coefficients across continuous variables.
- **Findings:** Vehicle weight (curb-weight) and engine displacement (engine-size) exhibit strong negative correlations with fuel efficiency variables (city-mpg and highway-mpg).



4.5 Fuel Efficiency by Manufacturer

- **Method:** Combined fuel efficiency metric calculated as $\text{average-mpg} = (\text{city-mpg} + \text{highway-mpg}) / 2$ and grouped by manufacturer.
- **Findings:** **Chevrolet** achieves the highest average fuel efficiency (~43.5 MPG). High-performance brands (Jaguar, BMW, Porsche) trade efficiency for increased horsepower and displacement.



5. Summary Metrics & Key Findings

Metric / Query	Finding	Context / Details
Top 5 Most Expensive Vehicles	1. BMW (\$41,315) 2. Mercedes-Benz (\$40,960) 3. Porsche (\$37,028) 4. BMW (\$36,880) 5. Jaguar (\$36,000)	Luxury performance segment driving maximum price bounds.
Most Fuel-Efficient Make	Chevrolet	Leading combined fuel economy (~43.5 MPG).
Largest Engine Displacement	Jaguar	326 CID (12-cylinder engine architecture).
Most Represented Make	Toyota	32 total vehicle models in cleaned dataset.

6. Recommendations & Business Impact

- Market Segmentation Strategy:** Dealerships and manufacturers targeting budget-conscious consumers should prioritize hatchback and entry-level sedan body styles with smaller displacement engines.

2. **Efficiency Trade-off Management:** Fleet operators seeking cost efficiency should focus on high-MPG manufacturers like Chevrolet, balancing curb weight against fuel mileage requirements.
3. **Pricing Model Optimization:** Linear regression modeling for vehicle appraisal should heavily weight horsepower, engine-size, and curb-weight as key predictor variables.

7. Conclusion

The EDA successfully highlights the major structural and performance drivers of automobile pricing and fuel economy. By eliminating missing observations and standardizing schema data types, the final dataset provides clear correlation evidence connecting engine displacement directly to market value and inversely to fuel economy.